Application of new braided soft actuator design in biomimetic robot locomotion

APPLICATION OF NEW BRAIDED SOFT ACTUATOR DESIGN IN BIOMIMETIC ROBOT LOCOMOTION

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This paper presents applications of novel soft actuator design; a combination of different braided angles of artificial muscle actuator in biomimetic locomotion. The actuator construction is based on the integration of artificial muscle contraction and extension theory. The actuator comprises of fiber-reinforced inside silicone rubber and is capable to create extension, contraction and one-sided bending motion. Based on the experimental results, the developed soft actuator can realize leech, swimming-frog and inchworm-like locomotion behavior after standard air pressure driving experiment is executed.

1. Introduction

Actuators have been vigorously utilized in many applications in modern society these days. The need of soft-surfaced actuator that can produce soft movement is increasing, as researchers are focusing on potential research where soft actuator could be useful in producing higher flexibility in its movement in various applications of robotics and automation. Existing developed soft actuator have shown promises in producing soft movements to objects and human [1-4].

Soft actuator actuation principle of extension and contraction can be achieved by applying different actuator design as developed by the previous researchers which includes chambers [1], fiber reinforced [1-3], and shape-type [4]. For the fiber reinforced design, the fiber may be braided by having certain degrees of angles or just in circular direction. This type of actuators is used by researchers in many fields of studies. A device utilizing soft actuator with braided angle technology has been developed to help doctors in performing colonoscopy insertion by changing its stiffness. The technology is used to reduce collision and prevent injuries during insertion process. The soft actuator will only change its stiffness and limit its function from extend, contract or bend when the air is supplied [2].

The braided angle type actuator also has been used in development of combining the contraction and extension principle in realizing the one-sided bending motion [3].

Recently, animals locomotion actively being imitate in building a robot which referred as biomimetic. The exploration in the biomimetic robotics field has produced a soft starfish robot constructed using pneumatic balloon channels. The actuator will bend due to inflation of the balloons thus mimics the starfish locomotion [4]. There is also a swimming robot that mimicked the swimming locomotion of manta stingrays. The manta swimming robot used a circular braided actuator with two internal chambers that enable it to bend in both sides [1].

In this paper, we developed a new braided angle soft actuator design in mimicking leech, swimming frog-like and inchworm locomotion. The purpose of this study is to broaden the application for the braided angle type actuator in the biomimetic locomotion.

2. Braided Angles and Its Application on the Developed Actuators

Study conducted by Iwata *et al.* [3] has proven that a single chamber hydraulic actuator knitted with fiber of angle 23.5° around it will lead to contraction and angle of 66.5° will result in extension of the actuator length. The attachment of both hydraulic actuators; contracted and extended actuators, as shown in Fig. 1 has made both actuators bended to the side of contracted actuator.

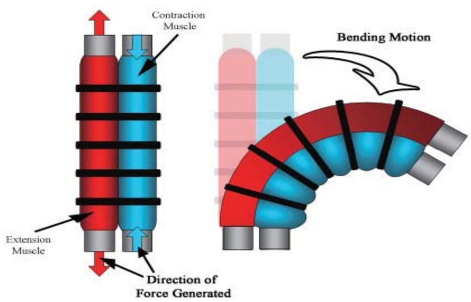

Figure 1. Generation of bending motion to the contracted side when actuators with extension muscle and contraction muscle are banded together [3].

Our pneumatic soft actuator design was inspired by his finding but we enforced different material of fibers with two angles knitted differently at each half of a single chamber actuator where (θ_1) on the top side and (θ_2) on the bottom side; as demonstrated in Fig. 2. We fixed the inside and outside diameter, thickness and length of the soft actuator for all models to be 8mm, 14mm, 3mm and 148mm respectively.

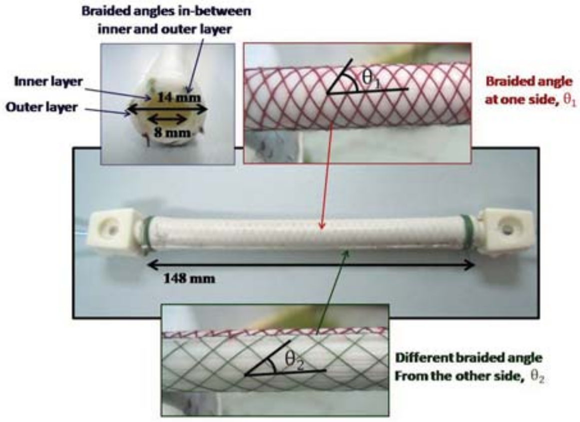

Figure 2. A single chamber actuator, braided with two different angles.

3. Fabrication Process

The developed pneumatic soft actuator was designed using 3D-CAD software INVENTOR^{TM} from Autodesk. The molds designed are used to construct and give the shape of the soft actuator. The molds will then be fabricated by CNC machine based on the 3D model designed.

Figure 3 and 4 shows the head and tail part of actuator with vacuum suction cup to control the locomotion of the actuator. The head has two insertion tubes, one for supplying air pressure into the inner chamber and the other one is for vacuum suction while the tail has only one tube connected to it which is the vacuum suction.

We have fabricated 6 models of actuators namely *Model A, B, C1, C2, D* and *E*. All actuator models have the same basic construction consist of head, tail and body parts. The construction of the actuator body part comprises of three layers. The first inner layer was fabricated beforehand by using silicone rubber material mixed with curing agent.

Vacuum tube

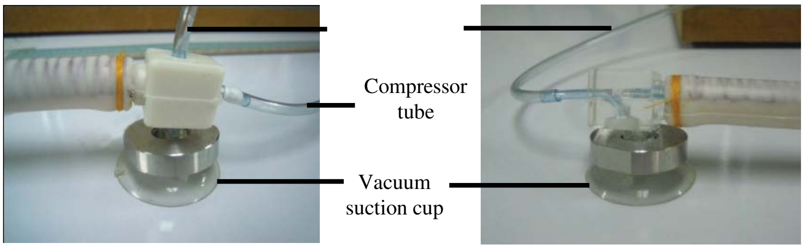

Figure 3. Head part.

Figure 4. Tail part.

A rod as in Fig. 5 is placed on top of the mixture that has been poured into the mold before the upper and bottom molds are attached together. This rod will be placed in the center of molds to create a chamber to actuate the actuator. We used 2 molds of inner layer and outer layer as shown in Fig. 6.

Figure 5. Rod placed in the center of molds

The following process is to knit the fabricated inner layer with reinforced fiber. Our models differ by these knitting angles. Next, the outer rubber layer will cover the inner layer with the knitted fiber. For *Model C2*, we used other types of rubber material mixture to compare actuator from which material could produce better bending ability.

Figure 6. Upper and bottom parts of molds for inner layer.

Figure 7. Upper and bottom parts of molds for outer layer.

The mixture is heated in oven at temperature of 80°C for about 40 minutes. The heating process will fasten the curing process. It could take 24 hours for the mixture to cure without the heating process.

3.1. *Experimental Testing*

Experimental setup as in Fig. 8 was prepared. Two valves are used; one to actuate the motion form air compressor and the other to control the movement using vacuum pump. The experiment was conducted to validate whether the developed actuators could locomote as predicted or not. The efficiency of movement based on the maximum displacement they can achieve at every cycle of locomotion can also be evaluated. Based on the result, it is proven that the developed actuators are able to locomote and the movement appears to be similar like a leech, a swimming frog and an inchworm.

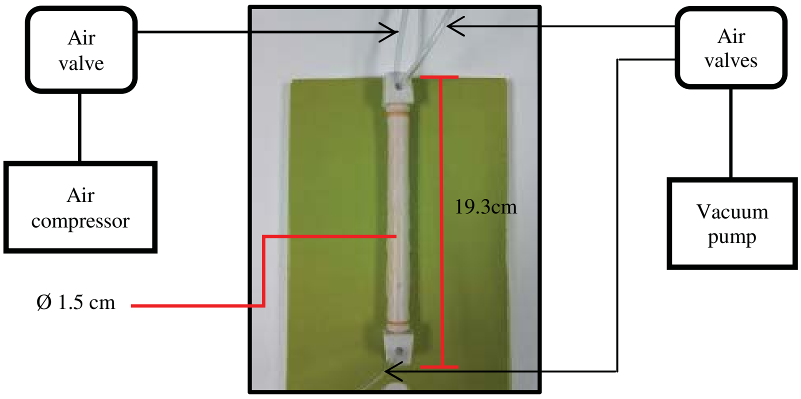

Figure 8. Experimental setup.

3.2. *Actuator Locomotion*

Fig. 9 to 14 shows the motion of soft actuator before and after pressurized with air in one cycle of locomotion. The air pressure and vacuum pressure applied to all models was 0.15MPa and -0.4MPa respectively.

Model A is able to elongate its body due to extension of fiber-reinforced in axial direction when body chamber is pressurized while *Model B* can shorten its body because of contraction of whole fiber-reinforced in the body. For *Model C1* and *C2*, the side with knitted $\theta_1°$ is contracted while the other side $\theta_2°$ is extended; thus producing bending motion to the contracted side.

Model C1 and *C2* use the same angles with different mixing material. Silicon rubber used in *Model C1* is from Dow Corning Corporation, using Silastic P-1 type (as base) and curing agent while *Model C2* is made from Shin Etsu Silicones Corporation using room temperature vulcanizing type (KE-1603-A-B). In the experiment conducted, the actuators designed for *Model C1* and *C2* produced bending angle 32^o and 31.5^o respectively. We found that both rubber materials give not much different in the deformation.

Some optimization in the bend-type actuator used in mimicking swimming frog has been made, where the actuator can bend up to 105^o. We tested the optimized version of bend type actuator in mimicking swimming frog, as in Fig. 13. Finally we tested the same actuator for inchworm locomotion as in Fig. 14.

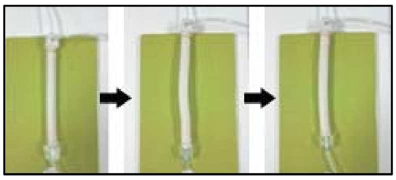

Figure 9. *Model A*, leech-like locomotion (extension).

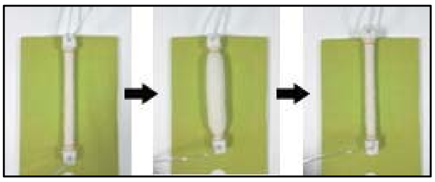

Figure 10. *Model B*, leech-like locomotion (contraction).

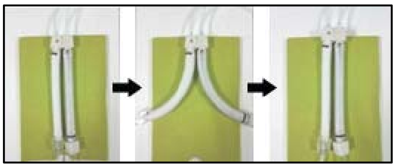

Figure 11. *Model C1*, swimming frog-like locomotion (extension and contraction).

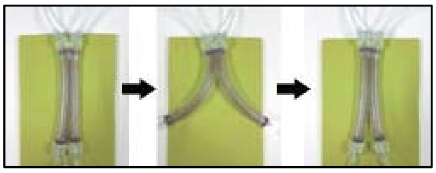

Figure 12. *Model C2*, swimming frog-like locomotion (extension and contraction).

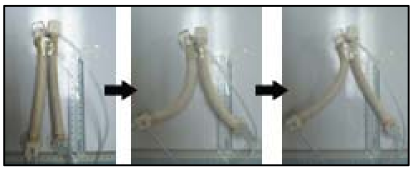

Figure 13. *Model D*, swimming frog-like locomotion (extension and contraction).

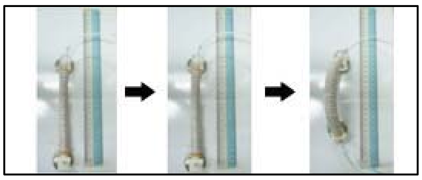

Figure 14. *Model E*, inchworm locomotion (extension and contraction).

All robots start to move by sucking air under its head to make sure the position of the head is fixed. Then air pressure is applied inside the robots body producing extension, contraction or bending motion. Next, the tail will suck air under its tail to fix the position of the tail. After that, the head and the pressure inside the body will released its suctions and pressure to produce movement.

Table 1 shows the maximum forward movement (displacement) of 6 robots models at every cycle of locomotion. The maximum displacement in 1 cycle of locomotion is achieved by *Model E*, the inchworm robot followed by *Model A, D, B, C1* and *C2*.

Table 1. Maximum forward displacement of 6 actuator models at every cycle of locomotion.

ACTUATOR MODEL	$\theta_1(°)$	$\theta_2(°)$	MOVEMENT GENERATED	DISPLACEMENT (cm/cycle)
A	78	78	Elongate (leech)	0.9
B	14	14	Shorten (leech)	0.5
C1	73	52	Bend	0.3
C2	73	52	Bend	0.2
D (optimized model from *C*)	84	52	Bend (swimming frog)	0.68
E	84	52	Bend (inchworm)	6.2

Selected robot of *Model A, B, D* and *E* were tested to know the fastest walking robot. Speed test was conducted at walking distance of 30 cm and the result is illustrated in Table 2. The fastest robot is achieved by *Model E*, inchworm robot.

Table 2. Speed test result at a fixed distance, 30cm.

ACTUATOR MODEL	DISTANCE (cm)	TIME (s)	SPEED (cm/s)
E	30	18	1.67
A	30	39	0.77
D	30	137	0.22
B	30	142	0.21

4. Discussion

Based on our experiment, we observed that *Model E* produced highest speed followed by *Model A, D* and *B*. *Model A* and *B* showing slower performance because of its structure produces less elongation and contraction compare to others and this could lead to slower performance.

Model D and *E* use the same braided angle. Even though the actuators of both models are same, *Model E* of inchworm robot can generate higher speed than *Model D* of swimming frog robot. This might due to high friction resulting from the robot body to the floor. *Model D* has larger area that is directly in contact with the floor compared to *Model E* where this could lead to slower performance.

In the future, the bending angle of the bend-type actuator can be further improve to generate greater displacement than the contracted and extended type actuator so that the robot will move faster like real animal. We also found that the actuator made from rubber material that have similar characteristic with silicon mixture yields better deformation.

5. Conclusion

In this research, we realized 6 models of braided type soft actuator which are able to generate movement based on the principle of extension and contraction. The technique of knitting fiber in actuator give benefit to the actuator's bending motion. With this ability, the actuator is found to be suitable in biomimetic robot locomotion application.

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